

30 Supergraphs

30.2 Superpropagators

Usually the propagator can be obtained directly from the part of the action of second order in the fields. If we write this quadratic part for a complex field ϕ_i (where i is a compound index including spacetime coordinates as well as spin and species labels) in the form

$$I_{\text{quad}}[\phi] = - \sum_{ij} D_{ij} \phi_i^* \phi_j, \quad (30.2.1)$$

with D_{ij} Hermitian, then as explained in Section 9.4 the propagator is simply $\Delta = D^{-1}$. The problem with this arises when the quadratic part of the action is invariant under a (linearized) gauge transformation

$$\phi_i \longrightarrow \phi_i + \xi_i, \quad (30.2.2)$$

for some class of ‘vectors’ ξ . In this case we have

$$\sum_i D_{ij} \xi_j = 0, \quad (30.2.3)$$

and we clearly cannot invert the ‘matrix’ D_{ij} . In electrodynamics the problem arises because the Lagrangian density is invariant under a gauge transformation $A_\mu \longrightarrow A_\mu + \partial_\mu \Lambda$. We have this problem here too: since the action is actually a functional of $\mathcal{D}_R^2 S_n$ and not of S_n itself, it is invariant under the transformation

$$S_n \longrightarrow S_n + \mathcal{D}_R X_n, \quad (30.2.4)$$

for any superfields X_n .

In the electrodynamics of charged particles the problems introduced by gauge invariance are typically handled by choosing a gauge, for instance by the Faddeev–Popov–de Witt method described in Section 15.5. But the problems raised here by invariance under the transformation (30.2.2) are much more like the problems of the effective field theory of photons at energies below the threshold for producing charged particles, where the theory is gauge-invariant simply because the action only involves gauge-invariant fields. In such theories there is a simpler option. In addition to the eigenvector ξ_i of D_{ij} with eigenvalue zero (taken for simplicity to be in a unique direction) we can find a set of orthonormal eigenvectors $u_{\nu i}$ with eigenvalues $d_\nu \neq 0$:

$$\sum_j D_{ij} u_{\nu j} = d_\nu u_{\nu i}, \quad \sum_i u_{\nu i}^* u_{\nu' i} = \delta_{\nu \nu'}, \quad \sum_i u_{\nu i}^* \xi_i = 0. \quad (30.2.5)$$

We can introduce a new set of integration variables ϕ' and ϕ'_ν :

$$\phi_i = \phi' \xi_i + \sum_\nu \phi'_\nu u_{\nu i}. \quad (30.2.6)$$

The sort of integral encountered in Feynman diagram calculations of quantum expectation values may then be written

$$\begin{aligned}
& \int \left[\prod_i d\phi_i d\phi_i^* \right] \exp\{iI_{\text{quad}}[\phi]\} \phi_a \cdots \phi_b^* \cdots \\
&= \mathcal{J} \int d\phi' d\phi'^* \int \left[\prod_\nu d\phi'_\nu d\phi'^*_\nu \right] \exp \left\{ -i \sum_\nu d_\nu |\phi'_\nu|^2 \right\} \\
& \quad \cdot \left[\phi'_a \xi_a + \sum_\nu \phi'_\nu u_{\nu a} \right] \cdots \left[\phi'_b \xi_b + \sum_\nu \phi'_\nu u_{\nu b} \right]^* \cdots, \tag{30.2.7}
\end{aligned}$$

where $\mathcal{J}$ is the Jacobian of the transformation (30.2.6). The integral over ϕ' and ϕ'^* is of course ill defined, because they do not appear in the argument of the exponential. But this doesn't matter if the action only involves gauge-invariant fields, because then ϕ_a, ϕ_b , etc. will be contracted with 'currents' J_a, J_b , etc. for which

$$\sum_a \xi_a J_a = 0. \tag{30.2.8}$$

We may thus write Eq. (30.2.7) as

$$\begin{aligned}
& \int \left[\prod_i d\phi_i d\phi_i^* \right] \exp\{iI_{\text{quad}}[\phi]\} \phi_a \cdots \phi_b^* \cdots \\
&= \mathcal{C} \int \left[\prod_\nu d\phi'_\nu d\phi'^*_\nu \right] \exp \left\{ -i \sum_\nu d_\nu |\phi'_\nu|^2 \right\} \left[\sum_\nu \phi'_\nu u_{\nu a} \right] \cdots \left[\sum_\nu \phi'_\nu u_{\nu b} \right]^* \cdots \\
& \quad + \xi\text{-terms}, \tag{30.2.9}
\end{aligned}$$

where ' ξ -terms' denotes terms proportional to one or more factors of ξ_a, ξ_b , etc., which vanish when contracted with J s satisfying Eq. (30.2.8), and $\mathcal{C}$ is the infinite constant $\mathcal{J} \int d\phi'$. The integral over the ϕ'_ν then gives

$$\begin{aligned}
& \int \left[\prod_i d\phi_i d\phi_i^* \right] \exp\{iI_{\text{quad}}[\phi]\} \phi_a \cdots \phi_b^* \cdots \\
& \propto \sum_{\text{pairings}} [-i\Delta_{ab}] \cdots + \xi\text{-terms}, \tag{30.2.10}
\end{aligned}$$

where the sum over pairings is over all ways of pairing indices on the ϕ s with indices on the ϕ^* s, and Δ_{ab} is the propagator

$$\Delta_{ab} = \sum_\nu \frac{u_{\nu a} u_{\nu b}^*}{d_\nu}. \tag{30.2.11}$$

Instead of actually evaluating the sum (30.2.11), we can use its defining property that

$$\sum_c D_{ac} \Delta_{cb} = \sum_\nu u_{\nu a} u_{\nu b}^* \equiv \Pi_{ab}, \tag{30.2.12}$$

with Π the projection operator on the space orthogonal to ξ :

$$\Pi^2 = \Pi, \quad \Pi\xi = 0. \quad (30.2.13)$$

The solution of Eq. (30.2.12) is unique only up to ξ -terms, but these do not matter if the fields ϕ_i appear in the action only in gauge-invariant combinations.

For instance, in electrodynamics, we can write the kinematic part of the action as

$$I_{\text{quad}}[A] = -\frac{1}{4} \int d^4x f_{\mu\nu} f^{\mu\nu} = +\frac{1}{2} \int d^4x A^\mu (\square \delta_\mu^\nu - \partial_\mu \partial^\nu) A_\nu.$$

The differential operator $-\square \delta_\mu^\nu + \partial^\nu \partial_\mu$ is not invertible, because it has a zero eigenvalue with eigenvectors of the form $\xi^\mu = \partial^\mu \Lambda$. The projection matrix onto the space orthogonal to these vectors is

$$\Pi_\mu{}^\nu(x, y) = [\delta_\mu^\nu - \partial_\mu \partial^\nu \square^{-1}] \delta^4(x - y),$$

where $\square^{-1} \delta^4(x - y)$ is any solution of the equation $\square[\square^{-1} \delta^4(x - y)] = \delta^4(x - y)$. The defining equation for the propagator is

$$(-\square \delta_\mu^\lambda + \partial_\mu \partial^\lambda) \Delta_\lambda{}^\nu(x, y) = \Pi_\mu{}^\nu(x, y),$$

with the solution

$$\Delta_\lambda{}^\nu(x, y) = \delta_\mu^\nu \Delta_F(x - y) + \partial_\mu \partial^\nu \text{-terms},$$

where $\Delta_F(x - y)$ is the usual Feynman propagator (6.2.16), satisfying $\square \Delta_F(x - y) = -\delta^4(x - y)$. (The factor 1/2 in the action does not appear in the defining equation for the propagator here because A_μ is a real field. The origin of the $-i\epsilon$ in the denominator of the Fourier integral in Eq. (6.2.16) is explained in the path-integral formalism in Section 9.2.)

Inspection of the first term in Eq. (30.1.9) shows that the defining equation for the superpropagator of the potential superfields is

$$\frac{1}{4} \mathcal{D}_L^2 \mathcal{D}_R^2 \Delta_{nm}^S(x, \theta; x', \theta') = \mathcal{P} \delta^4(x - x') \delta^4(\theta - \theta') \delta_{nm}, \quad (30.2.14)$$

where $\mathcal{P}$ is a superspace differential operator satisfying the conditions for a projection operator

$$\mathcal{P}^2 = \mathcal{P}, \quad \mathcal{P} \mathcal{D}_R = 0, \quad (30.2.15)$$

and $\delta^4(\theta - \theta')$ is the fermionic delta function introduced in Eq. (26.6.8). The solution is

$$\mathcal{P} = -\frac{1}{16\square} \mathcal{D}_L^2 \mathcal{D}_R^2. \quad (30.2.16)$$

(It is obvious that $\mathcal{P} \mathcal{D}_R = 0$. To check that $\mathcal{P}^2 = \mathcal{P}$, we need to use Eq. (26.6.12), which shows that $\mathcal{D}_R^2 \mathcal{D}_L^2 \mathcal{D}_R^2 = -16\square \mathcal{D}_R^2$.) The solution of Eq. (30.2.14) is

$$\begin{aligned} \Delta_{nm}^S(x, \theta; x', \theta') &= -\frac{1}{4\square} \delta^4(x - x') \delta^4(\theta - \theta') \delta_{nm} \\ &= \frac{1}{4} \Delta_F(x - x') \delta^4(\theta - \theta') \delta_{nm} + \mathcal{D}_R \text{-terms}. \end{aligned} \quad (30.2.17)$$

This is the superpropagator for a line created by a potential superfield $S_m^*(x', \theta')$ and destroyed by a potential superfield $S_n(x, \theta)$, which we must use in evaluating supergraphs for the action (30.1.9). To make contact with ordinary propagators, it is also of some interest to consider the superpropagator of the line created by a left-chiral superfield $\Phi_m^*(x', \theta')$ and destroyed by a left-chiral superfield $\Phi_n(x, \theta)$. These chiral superfields are obtained by operating on $S_n(x, \theta)$ with $\mathcal{D}_R^2$ and on $S_m^*(x', \theta')$ with $(\mathcal{D}_R^2)^* = \mathcal{D}_L'^2$, so the propagator for the left-chiral superfields is

$$\Delta_{nm}^\Phi(x, \theta; x', \theta') = \frac{1}{4} \mathcal{D}_R^2 \mathcal{D}_L'^2 \Delta_F(x - x') \delta^4(\theta - \theta') \delta_{nm}. \quad (30.2.18)$$

For instance, the term in the superpropagator of zeroth order in θ and θ' is

$$[\Delta_{nm}^\Phi(x, \theta; x', \theta')] \big|_{\theta=\theta'=0} = \frac{1}{4} \left(\frac{\partial}{\partial \theta_R} \right)^2 \left(\frac{\partial}{\partial \theta'_L} \right)^2 \Delta_F(x - x') \delta^4(\theta - \theta') \delta_{nm}. \quad (30.2.19)$$

To evaluate this, we recall Eq. (26.6.8) for the fermionic delta function:

$$\delta^4(\theta - \theta') = \frac{1}{4} ((\theta_L - \theta'_L)^T \epsilon (\theta_L - \theta'_L)) ((\theta_R - \theta'_R)^T \epsilon (\theta_R - \theta'_R)),$$

from which we find $(\partial/\partial \theta_R)^2 (\partial/\partial \theta'_L)^2 \delta^4(\theta - \theta') = 4$. Eq. (30.2.19) therefore yields

$$[\Delta_{nm}^\Phi(x, \theta; x', \theta')] \big|_{\theta=\theta'=0} = \Delta_F(x - x') \delta_{nm}, \quad (30.2.20)$$

which is just the usual propagator for the scalar components of the superfield.